Project: Implementing Grover's Algorithm in Qiskit

Objective

In this project you will implement Grover's search algorithm using Qiskit to locate marked elements in a large search space. You must build the Grover circuit entirely using Qiskit quantum gates, including the oracle, diffusion operator, Grover iterations, and measurement.
The final program must run and display a measurement histogram where the marked states appear with the highest probability.

References

IBM Quantum Learning Module:
https://quantum.cloud.ibm.com/learning/en/modules/computer-science/grovers

Lecture slides and notes provided in class.

Problem Setup

Consider a 10-qubit search space. This corresponds to $2^{10} = 1024$ possible bit strings.

Define a function f(x) such that f(x)=1 for two specific bit strings and f(x)=0 for all other inputs.

For this project use the following marked states:

0110011010
1101010001

Your Grover circuit must amplify the probability of measuring these two states.

Tasks

1. Create the Superposition State

Initialize the 10-qubit register and apply Hadamard gates to all qubits to create an equal superposition of all states.

2. Implement the Oracle

Construct an oracle circuit that flips the phase of the two marked states 0110011010 and 1101010001. The oracle must be implemented using Qiskit quantum gates.

The oracle should work for any two bit strings defined in the program. For example:
targets = ["0110011010","1101010001"]

Changing these values should automatically update the oracle.

3. Implement the Diffusion Operator

Construct the Grover diffusion operator.
Hint: The diffusion operator can be implemented using the structure $H \rightarrow U_0 \rightarrow H$.
Here U_0 performs a phase flip on all computational basis states except the $|000...0\rangle$ state. In other words, $|000...0\rangle$ remains unchanged while all other states receive a phase of -1.
You may implement U_0 using multi-controlled quantum gates available in Qiskit.

4. Perform Grover Iterations

Apply the Grover iteration multiple times. Run the circuit for several iteration counts such as 1 iteration, 3 iterations, 5 iterations, and 10 iterations. Observe how the probability distribution changes.

5. Execute the Circuit

Run the circuit using the Qiskit simulator. Use at least 1024 shots. Display the measurement histogram.
The histogram should clearly show the two marked states appearing with the highest probability.

Implementation Requirements

You must not use built-in Grover implementations provided by Qiskit such as qiskit.algorithms.Grover. The Grover circuit must be constructed manually using quantum gates.

Your code must clearly separate the following components:
- oracle construction
- diffusion operator
- Grover iteration
- main program that runs the circuit

Expected Output

When the program is executed:
1. The Grover circuit should run successfully.
2. A histogram of measurement results should appear.
3. The two marked states

0110011010
1101010001

should appear with the highest probability in the histogram.

Submission Instructions

Students must submit only a GitHub repository link.

The repository must contain:
• the complete Qiskit code (.py or .ipynb)
• a README file explaining how to run the program

When the TA runs the code, the Grover circuit should execute and the histogram output should appear. No written report is required.